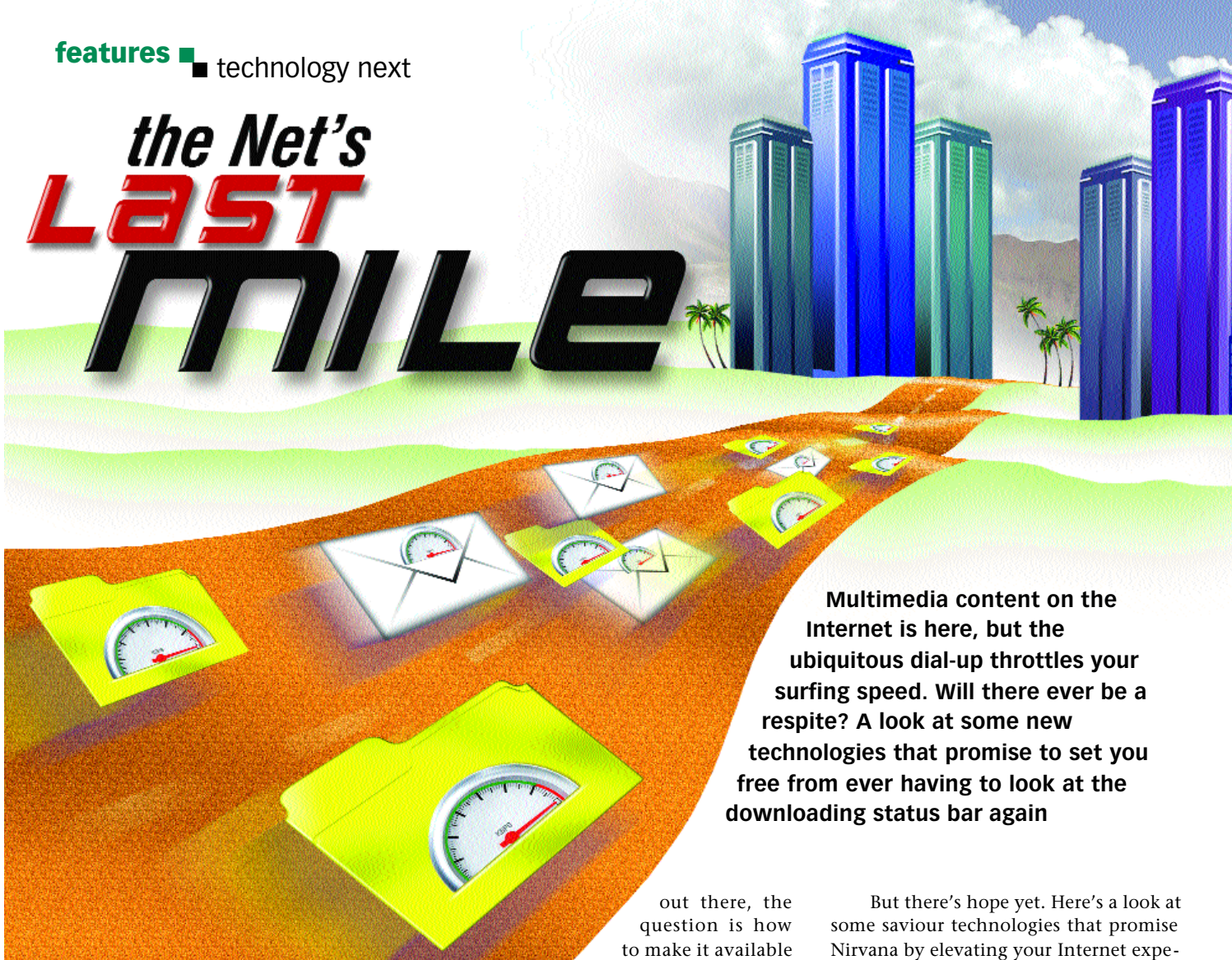


the Net's **LAST** MILE



Multimedia content on the Internet is here, but the ubiquitous dial-up throttles your surfing speed. Will there ever be a respite? A look at some new technologies that promise to set you free from ever having to look at the downloading status bar again

The Net today offers very vivid content such as animation-rich Flash Web sites, streaming audio and video news and full-motion videoconferencing with friends and family abroad. But these services require a high-bandwidth connection—something that every Net junkie dreams of.

The Internet actually operates on very high bandwidth pipelines, but the problem is that most of us never ever get to experience this speed. Take, for example, the case of the backbones of the Internet that are used to connect countries to one another. These are based on ultra high-speed optical connections such as OC-12 lines, which offer a bandwidth of 622.08 million bits per second—with such bandwidth at your disposal, you could transfer an entire CD in 9 seconds flat, twice as fast as the practical data transfer rates of most hard disk drives used in PCs!

With all this delicious bandwidth

out there, the question is how to make it available to end-users as efficiently and cheaply as possible. The popular standard of Internet access for most surfers today is through the trusty but archaic telephone network. Dial-up connections provide cheap access to the Net but prove to be a bottleneck for bandwidth-demanding applications such as real-time audio and video applications, as they are only capable of delivering transfer rates at a maximum of 6 Kbps. With the proliferation of peer-to-peer applications, millions of files are just asking to be downloaded, and this last mile connectivity is the only hindrance. Other alternatives like DSL and cable modem are too expensive and their availability is limited. Additionally, the effective bandwidth of these supposedly 'broadband' services is curbed in most cases and the end-users are left with a rather narrow-band experience.

But there's hope yet. Here's a look at some saviour technologies that promise Nirvana by elevating your Internet experience to a higher plane.

Internet at the speed of light

Very few establishments, much less homes, are located within close proximity of an Internet backbone. So far, the only way to create a high-bandwidth link with the Internet backbone was by setting up optical fibre connections, but this is not a viable option for most. One of the most promising technologies that effectively deal with this problem is Free Space Optics (FSO). FSO uses laser beams capable of carrying enormous quantities of data at high speeds. These laser devices can be mounted on top of buildings pointing at one another, creating an invisible laser mesh of optical connectivity over short distances.

The US Army has been using this technology since the 1980s to carry digital information through modulated laser beams. However, →

Cost and time to install per building

Fibre optics link: up to Rs 1 crore; four to 12 months
FSO link: about Rs 10 lakh; two to three days